

# Explanation of Section 7: Completeness of the Three-Flavour Invariant Entropy-Maximal Locus

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## 1 Purpose of Section 7

The goal of Section 7 is to justify a completeness claim. Earlier sections show that the two-parameter cubic/tetrahedral plane

$$\lambda_{iiii} = u, \quad \lambda_{iijj} = w \quad (i \neq j), \quad \text{all other components} = 0 \quad (1)$$

lies inside the one-leg entropy-maximal locus and is closed under the one-loop RG flow. Section 7 asks whether this plane is merely an example, or whether it is the whole answer.

The answer claimed in the note is:

For real three-flavour quartic tensors, the full one-loop RG-invariant locus inside the 1|3 entropy-maximal manifold is, modulo flavour rotations, exactly the cubic/tetrahedral plane (1).

Equivalently, every tensor in the invariant entropy-maximal locus can be written as

$$\lambda_{ijkl} = w D_{ijkl} + (u - 3w) \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a, \quad (2)$$

where  $\{n^1, n^2, n^3\}$  is an orthonormal frame and

$$D_{ijkl} := \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}. \quad (3)$$

The special line  $u = 3w$  is the  $\text{SO}(3)$ -symmetric line. Points with  $u \neq 3w$  are generally only cubic/tetrahedral invariant, not fully rotationally invariant.

The phrase “modulo rotations” means that if a tensor is brought to the form (1) by an  $\text{O}(3)$  change of flavour basis, it is counted as the same type of solution. The RG equations and the entropy condition are  $\text{O}(3)$ -equivariant, so rotating a solution gives another solution.

## 2 The Objects Being Studied

We study a real fully symmetric quartic tensor

$$\lambda_{ijkl} = \lambda_{(ijkl)}, \quad i, j, k, l = 1, 2, 3. \quad (4)$$

It has

$$\dim \text{Sym}^4(\mathbb{R}^3) = \binom{3+4-1}{4} = 15 \quad (5)$$

independent real components.

The unnormalized one-leg reduced density matrix is

$$R_{ij} := \sum_{k,l,m=1}^3 \lambda_{iklm} \lambda_{jklm}. \quad (6)$$

The normalized 1|3 entropy is maximal precisely when  $R$  is proportional to the identity:

$$R_{ij} = \frac{1}{3}(\text{tr} R) \delta_{ij}. \quad (7)$$

Equivalently, the traceless symmetric part of  $R$  vanishes:

$$T_{ij} := R_{ij} - \frac{1}{3}(\text{tr} R) \delta_{ij} = 0. \quad (8)$$

This gives five independent equations, because a traceless symmetric  $3 \times 3$  matrix has five independent components.

The one-loop beta function is

$$\dot{\lambda}_{ijkl} = -\epsilon \lambda_{ijkl} + \kappa (\lambda_{ijmn} \lambda_{mnkl} + \lambda_{ikmn} \lambda_{mnjl} + \lambda_{ilmn} \lambda_{mnjk}). \quad (9)$$

For notational simplicity, the attached note sets  $\epsilon = 1$  in several places. This does not affect the representation-theoretic discussion.

The invariant entropy-maximal locus is not just the set satisfying  $T = 0$ . We require the RG trajectory to remain inside  $T = 0$ . A standard way to impose that condition algebraically is the Lie-chain condition

$$T = 0, \quad \mathcal{L}_\beta T = 0, \quad \mathcal{L}_\beta^2 T = 0, \quad \dots \quad (10)$$

Here  $\mathcal{L}_\beta$  means differentiation along the beta-function vector field:

$$\mathcal{L}_\beta F := \sum_a \beta_a \frac{\partial F}{\partial \lambda_a}. \quad (11)$$

If  $T = 0$  but  $\mathcal{L}_\beta T \neq 0$ , the RG flow immediately leaves the entropy-maximal manifold. If  $T = \mathcal{L}_\beta T = 0$  but some higher Lie derivative is nonzero, the flow leaves at higher order in RG time.

### 3 Representation-Theory Decomposition

The crucial simplifying fact is that  $\text{Sym}^4(\mathbb{R}^3)$  decomposes into irreducible  $\text{SO}(3)$  representations as

$$\text{Sym}^4(\mathbb{R}^3) = V_0 \oplus V_2 \oplus V_4, \quad 15 = 1 + 5 + 9. \quad (12)$$

The labels 0, 2, 4 are angular momenta. Thus:

- $V_0$  is the scalar or singlet piece.
- $V_2$  is a spin-2 traceless symmetric tensor piece.

- $V_4$  is a spin-4 harmonic quartic piece.

The singlet is generated by

$$D_{ijkl} = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}. \quad (13)$$

A general quartic tensor can therefore be written schematically as

$$\lambda = sD + \lambda^{(2)} + H, \quad (14)$$

where  $sD \in V_0$ ,  $\lambda^{(2)} \in V_2$ , and  $H \in V_4$ .

The spin-2 part is detected by the trace matrix

$$\tau_{ij} := \lambda_{ijkk}. \quad (15)$$

Its traceless part is

$$q_{ij} := \tau_{ij} - \frac{1}{3}(\text{tr}\tau)\delta_{ij}. \quad (16)$$

The condition  $q = 0$  says that the  $V_2$  piece is absent. In components,  $q = 0$  becomes

$$A + D + F = D + K + M = F + M + P, \quad (17)$$

and

$$B + G + I = 0, \quad C + H + J = 0, \quad E + L + N = 0. \quad (18)$$

When  $q = 0$ , the tensor has the simpler form

$$\lambda = sD + H, \quad H_{ijkk} = 0. \quad (19)$$

The condition  $H_{ijkk} = 0$  means that  $H$  is a harmonic rank-four tensor. In polynomial language,  $H_{ijkl}x_i x_j x_k x_l$  is a harmonic homogeneous quartic polynomial: its Laplacian vanishes.

## 4 Step 1 of Section 7: RG Invariance Forces $q = 0$

The first substantive claim in Section 7 should be stated with the full RG-invariance condition, not merely first-order tangency. The correct implication is

$$T = 0, \quad \mathcal{L}_\beta T = 0 \quad \mathcal{L}_\beta^2 T = 0 \quad \mathcal{L}_\beta^3 T = 0 \quad \implies \quad q = 0 \quad \text{over the real locus.} \quad (20)$$

Equivalently, if a real tensor is 1|3 entropy-maximal and the one-loop RG trajectory through it remains inside the entropy-maximal manifold, then the tensor cannot contain a spin-2 trace component.

First-order tangency alone is not enough. There are real points with

$$T = 0, \quad \mathcal{L}_\beta T = 0, \quad q \neq 0, \quad (21)$$

but such points fail at a higher derivative and leave the maximal-entropy locus at higher order in RG time. This is why the invariant locus is defined by the full Lie chain, not just by tangent vectors at first order.

This is the key reason the problem simplifies. Before imposing (20), the tensor has 15 components. After imposing  $q = 0$ , it lies in

$$V_0 \oplus V_4, \quad (22)$$

which has only  $1 + 9 = 10$  dimensions.

Intuitively, the  $V_2$  component is an anisotropy that can hide at one instant inside the quadratic condition  $T = 0$ . Some exceptional  $q \neq 0$  points even have first-order tangency. But the nonlinear RG flow exposes the anisotropy in the Taylor expansion of  $T(t)$  along the RG trajectory. Requiring every Taylor coefficient of  $T(t)$  to vanish removes this possibility.

Here is the precise proof used in the computation. Work in the real polynomial ring generated by the 15 independent symmetric components

$$A, B, C, D, E, F, G, H, I, J, K, L, M, N, P. \quad (23)$$

Let

$$T_{ij}^{(r)} := \mathcal{L}_\beta^r T_{ij}, \quad r = 0, 1, 2, 3. \quad (24)$$

For each  $r$ , keep the five independent components

$$T_{11}^{(r)}, \quad T_{12}^{(r)}, \quad T_{13}^{(r)}, \quad T_{22}^{(r)}, \quad T_{23}^{(r)}, \quad (25)$$

with  $T_{33}^{(r)}$  determined by tracelessness. Let  $I_3$  be the ideal generated by these twenty polynomials. The claim (20) is equivalent to the statement that the real algebraic set of  $I_3$  has no nonzero point with  $q \neq 0$ .

Because this is a system of polynomial equations and inequalities over  $\mathbb{R}$ , it can be decided exactly by real quantifier elimination. The exact statement checked is

$$\exists \lambda \in \mathbb{R}^{15} : \quad T^{(r)} = 0 \ (r = 0, 1, 2, 3), \quad \|\lambda\|^2 > 0, \quad q_{ij}q_{ij} > 0. \quad (26)$$

Mathematica's exact real quantifier elimination returns **False** for this existential sentence. Since real quantifier elimination is a complete decision procedure for polynomial equations and inequalities over a real closed field, this is a proof, not a floating-point check.

The corresponding Mathematica skeleton is:

```
vars = {A,B,C,D,E,F,G,H,I,J,K,L,M,N,P};

Lie[expr_] := Sum[beta[[a]] D[expr, vars[[a]]], {a, Length[vars]}];

Tchain = Flatten@Table[
  independentTracelessComponents[Nest[Lie, T, r]],
  {r, 0, 3}
];

qnorm = Tr[q.q];
norm = Total[vars^2];

Resolve[
  Exists[vars,
    And @@ Thread[Tchain == 0] && norm > 0 && qnorm > 0
  ], Reals]
```

The output is

**False**

Thus every real nonzero RG-invariant point in the maximal  $1/3$  entropy locus has  $q = 0$ . This is the rigorous bridge from the full tensor problem to the scalar-plus-harmonic problem.

## 5 Step 2 of Section 7: The Harmonic Entropy Equation

Once  $q = 0$ , write

$$\lambda = sD + H, \quad H_{ijkk} = 0. \quad (27)$$

The singlet piece  $sD$  is rotationally invariant. It cannot contribute to the traceless spin-2 part of  $R_{ij}$  by itself. Moreover, the cross-term between  $sD$  and  $H$  vanishes in the relevant spin-2 channel because  $H$  is traceless. Thus the remaining entropy condition is a condition on  $H$  alone:

$$C_{ij} := H_{iklm}H_{jklm} - \frac{1}{3}H_{aklm}H_{aklm}\delta_{ij} = 0. \quad (28)$$

This says that the contraction

$$H_{iklm}H_{jklm} \quad (29)$$

must be isotropic. It must be proportional to  $\delta_{ij}$ .

Equation (28) is a system of five homogeneous quadratic equations on the 9-dimensional harmonic space  $V_4$ . The note claims that its nonzero real solutions form a single  $O(3)$  orbit, up to overall scale and sign. A representative is the cubic harmonic

$$H_{ijkl} = h \left( \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a - \frac{1}{5} D_{ijkl} \right). \quad (30)$$

Here  $\{n^1, n^2, n^3\}$  is an orthonormal frame. In the frame  $n^1 = (1, 0, 0)$ ,  $n^2 = (0, 1, 0)$ ,  $n^3 = (0, 0, 1)$ , the first term is the tensor corresponding to

$$x_1^4 + x_2^4 + x_3^4. \quad (31)$$

Subtracting  $D/5$  removes its trace.

The coefficient  $1/5$  is fixed as follows. First,

$$\left( \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a \right) \delta_{kl} = \sum_{a=1}^3 n_i^a n_j^a = \delta_{ij}. \quad (32)$$

Second,

$$D_{ijkk} = 5\delta_{ij} \quad (33)$$

in three dimensions. Therefore

$$\left( \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a - \frac{1}{5} D_{ijkl} \right) \delta_{kl} = 0. \quad (34)$$

So (30) is indeed harmonic.

The cubic harmonic also satisfies the needed isotropy property:

$$H_{iklm}H_{jklm} = \frac{2}{5}h^2\delta_{ij}. \quad (35)$$

Thus it solves (28).

The nontrivial completeness statement is that, over the real numbers, these are all the solutions of (28) modulo rotations and scaling.

## 6 Recovering the Cubic/Tetrahedral Plane

Combining the scalar and harmonic pieces gives

$$\lambda_{ijkl} = sD_{ijkl} + h \left( \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a - \frac{1}{5} D_{ijkl} \right). \quad (36)$$

Equivalently,

$$\lambda_{ijkl} = \left( s - \frac{h}{5} \right) D_{ijkl} + h \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a. \quad (37)$$

Comparing with (2), we identify

$$w = s - \frac{h}{5}, \quad u - 3w = h. \quad (38)$$

Equivalently,

$$u = 3s + \frac{2}{5}h, \quad w = s - \frac{1}{5}h. \quad (39)$$

In the aligned frame, this gives

$$\lambda_{iiii} = 3s + \frac{2}{5}h = u, \quad (40)$$

and for  $i \neq j$ ,

$$\lambda_{iijj} = s - \frac{1}{5}h = w, \quad (41)$$

with all components containing an odd number of any flavour equal to zero. Hence one recovers exactly the two-parameter cubic/tetrahedral plane.

The SO(3)-symmetric line is  $h = 0$ , equivalently

$$u = 3w. \quad (42)$$

The rest of the plane has only the discrete symmetry of a cube or tetrahedral arrangement, depending on convention, not the full SO(3) symmetry.

## 7 What Section 7 Proves and What It Assumes

The logical chain is

$$\begin{aligned} & \text{entropy-maximal and RG-invariant} \\ & \implies q = 0 \\ & \implies \lambda = sD + H \\ & \implies H = \text{cubic harmonic up to rotation} \\ & \implies \lambda = \text{cubic/tetrahedral plane up to rotation.} \end{aligned} \quad (43)$$

The conceptually important point is that Section 7 is not about finding fixed points. It is about finding the whole RG-invariant submanifold inside the 1/3 entropy-maximal locus. Once that invariant submanifold is found, Section 8 separately asks which points on it are actual fixed points, i.e. where  $\dot{u} = \dot{w} = 0$ .

The technically hardest parts are:

1. proving the Lie-chain implication  $T = \mathcal{L}_\beta T = \mathcal{L}_\beta^2 T = \mathcal{L}_\beta^3 T = 0 \Rightarrow q = 0$  over the real locus;
2. proving that (28) has only the cubic harmonic orbit.

The note states these as direct computational results. For a publication-level derivation, these should be supported by a reproducible computer algebra notebook or a more invariant analytic proof.

## A Group Theory Background

This appendix collects the minimum representation theory needed for Section 7.

### A.1 Rotations and Tensor Representations

The flavour group is  $O(3)$  or  $SO(3)$ , acting on the real vector space  $\mathbb{R}^3$ . A vector transforms as

$$v_i \mapsto O_{ij} v_j, \quad O \in O(3). \quad (44)$$

A rank-four tensor transforms as

$$\lambda_{ijkl} \mapsto O_{ia} O_{jb} O_{kc} O_{ld} \lambda_{abcd}. \quad (45)$$

Because the tensor is fully symmetric, it belongs to  $\text{Sym}^4(\mathbb{R}^3)$ .

An irreducible representation is a vector space of tensors that cannot be decomposed further into smaller rotation-invariant subspaces. For  $SO(3)$ , irreducible representations are labelled by angular momentum  $\ell = 0, 1, 2, \dots$  and have dimension  $2\ell + 1$ .

### A.2 Symmetric Tensors and Spherical Harmonics

A symmetric rank- $m$  tensor  $T_{i_1 \dots i_m}$  can be associated with a homogeneous polynomial

$$P_T(x) = T_{i_1 \dots i_m} x_{i_1} \cdots x_{i_m}. \quad (46)$$

For quartic tensors,  $m = 4$ . The polynomial  $P_T$  has degree four.

The Laplacian of this polynomial corresponds to tracing two indices of the tensor:

$$\nabla^2 P_T(x) \propto T_{iikl} x_k x_l. \quad (47)$$

Therefore a traceless symmetric tensor corresponds to a harmonic polynomial:

$$T_{iikl} = 0 \quad \Longleftrightarrow \quad \nabla^2 P_T = 0. \quad (48)$$

Harmonic homogeneous polynomials of degree  $\ell$  in three dimensions transform as the spin- $\ell$  representation of  $SO(3)$  and have dimension  $2\ell + 1$ .

For quartics, the highest harmonic part is spin 4, with dimension

$$2\ell + 1 = 9. \quad (49)$$

### A.3 Why $\text{Sym}^4(\mathbb{R}^3) = V_0 \oplus V_2 \oplus V_4$

Every homogeneous quartic polynomial can be decomposed into harmonic pieces multiplied by powers of  $r^2 = x_i x_i$ :

$$P_4(x) = H_4(x) + r^2 H_2(x) + r^4 H_0. \quad (50)$$

Here:

- $H_4$  is a degree-four harmonic polynomial, spin 4, dimension 9.
- $H_2$  is a degree-two harmonic polynomial, spin 2, dimension 5.
- $H_0$  is a constant, spin 0, dimension 1.

Therefore

$$\text{Sym}^4(\mathbb{R}^3) = V_4 \oplus V_2 \oplus V_0, \quad (51)$$

with dimensions

$$15 = 9 + 5 + 1. \quad (52)$$

The order is often written as  $V_0 \oplus V_2 \oplus V_4$  because the scalar part is conceptually simplest.

### A.4 The Scalar Tensor $D_{ijkl}$

The unique rotationally invariant quartic tensor is

$$D_{ijkl} = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}. \quad (53)$$

It corresponds to the polynomial

$$D_{ijkl}x_i x_j x_k x_l = 3(x_i x_i)^2 = 3r^4. \quad (54)$$

Thus  $D$  spans the spin-0 part  $V_0$ .

### A.5 The Spin-2 Part and the Trace Matrix

For a quartic tensor, tracing two indices gives

$$\tau_{ij} = \lambda_{ijkk}. \quad (55)$$

This is a symmetric rank-two tensor. Any symmetric rank-two tensor decomposes into a trace and a traceless part:

$$\tau_{ij} = \frac{1}{3}(\text{tr}\tau)\delta_{ij} + q_{ij}. \quad (56)$$

The trace part belongs to spin 0. The traceless part  $q_{ij}$  belongs to spin 2.

Thus  $q = 0$  is precisely the statement that the quartic tensor has no  $V_2$  component.



## A.6 The Cubic Harmonic

In the standard coordinate frame, the cubic quartic is

$$x_1^4 + x_2^4 + x_3^4. \quad (57)$$

This polynomial is invariant under the cubic symmetry group, but not under all rotations. Its harmonic projection is obtained by subtracting a multiple of  $r^4$ :

$$H_{\text{cub}}(x) = x_1^4 + x_2^4 + x_3^4 - \frac{3}{5}r^4. \quad (58)$$

In tensor language this is equivalent to

$$H_{ijkl} = h \left( \sum_{a=1}^3 n_i^a n_j^a n_k^a n_l^a - \frac{1}{5} D_{ijkl} \right). \quad (59)$$

The different factors of  $3/5$  and  $1/5$  arise because

$$D_{ijkl} x_i x_j x_k x_l = 3r^4. \quad (60)$$

The cubic harmonic is a spin-4 object. Rotating the orthonormal frame gives the full  $O(3)$  orbit of this harmonic.

## A.7 Why Orbit Classification Appears

The equations are rotationally invariant. Therefore if  $H$  solves the harmonic isotropy equation, so does every rotated tensor  $O \cdot H$ . The meaningful question is not how many individual tensors solve the equation, but how many inequivalent orbits solve it.

The claim in Section 7 is that, for  $N = 3$ , there is only one nonzero real orbit: the cubic harmonic orbit. This is special to three dimensions. In higher dimensions, analogous harmonic isotropy equations can have multiple inequivalent orbits.